



IMT Atlantique

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Improving Classification Accuracy with Graph Filtering

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Outline

- ① Deep Learning and Few-shot Learning
- ② Methodology
- ③ Experiments
- ④ Conclusion and Perspectives

Outline

① Deep Learning and Few-shot Learning

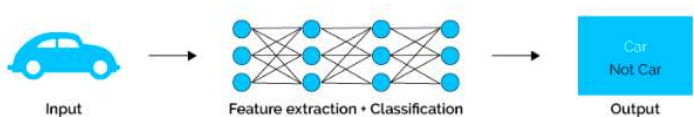
② Methodology

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Introduction

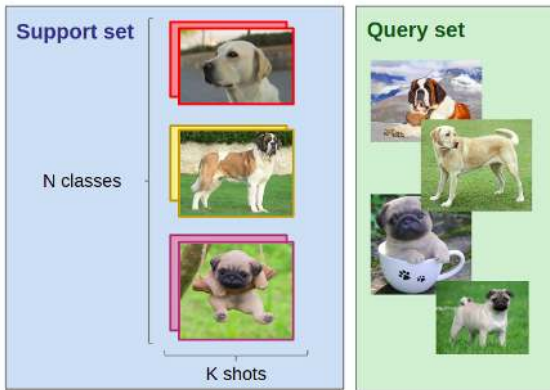
- Deep Learning is the state of the art for many machine learning problems



- Reaching peak performance typically requires large annotated datasets
- Performing predictions on batches of unlabeled inputs can boost performance: this setting is called *transductive*.

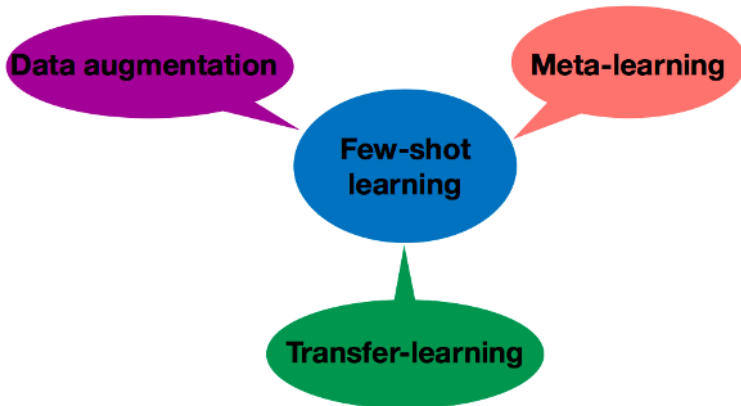
Transductive predictions are even more impactful in the case of few-shot learning

► N -way K -shot image classification



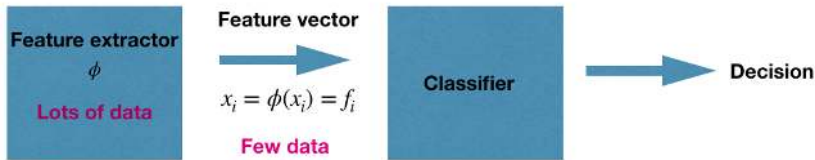
- For $K \leq 5$, classify the **query images** among the N classes given the $N \times K$ images in the **support set**

Few-shot Learning Methods



Main idea of the paper:

- **Idea:** Insert a denoising method after feature extraction and before classification after a model has been trained



- **Goal:** filter out the high frequencies within the feature representation of data to improve classification performance
- **Tool:** graph signal processing framework

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Graph Signal Processing (GSP)

► GSP allows to manipulate signals defined on **graph structures**

- Assume $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{W})$ is connected. If not, just process each component separately

$\mathcal{V} = \{ 1, 2, \dots, n \}$: finite set of n nodes or vertices

$\mathcal{E}$: set of edges

$\mathbf{W} : \mathcal{E} \rightarrow \mathbb{R}$: map from the set of edges to scalar values

- Assume undirected graph with fixed graph structure
- Vertex signal $\mathbf{f}_i : \mathcal{V} \rightarrow \mathbb{R}$
- Graph signal $\mathbf{f} = [\mathbf{f}_1, \mathbf{f}_2, \dots, \mathbf{f}_n]^\top$

► **Graph filters** are systems that take a **graph signal as an input** and **produce another signal** indexed by the same graph as the output

Class Graphs

- Define a **similarity matrix** $\mathbf{W}$ between **labeled** samples

$$\mathbf{W}_{i,j} = w(\mathbf{f}_i, \mathbf{f}_j) \text{ if } i \neq j \text{ and } 0 \text{ otherwise,}$$

where w is a similarity measure on tensors.

- This matrix defines a **graph** $\mathcal{G}$

$$\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{W})$$

with $\mathcal{V} = \{1, \dots, |\mathbf{f}|\}$.

- Generate the **adjacency matrix** $\mathbf{W}$ of $\mathcal{G}$ for each class using **k -nearest neighbors**

$$\mathbf{W}_{i,j} = \begin{cases} \mathbf{W}_{i,j} & \text{if } \mathbf{W}_{i,j} \text{ is among the } k \text{ largest entries of row,} \\ 0 & \text{otherwise.} \end{cases}$$

- Normalized Laplacian matrix of $\mathcal{G}$

$$\mathbf{L} = \mathbf{I} - \mathbf{D}^{-\frac{1}{2}} \mathbf{W} \mathbf{D}^{-\frac{1}{2}},$$

where $\mathbf{D}$ is the diagonal degree matrix.

Graph Filters

- ▶ Eigenvalues λ of $\mathbf{L}$ are interpreted as **frequencies**
- ▶ A **filter** is usually defined by its **spectral response**

$$h : \lambda \mapsto h(\lambda)$$

where $\lambda \in \lambda$.

- ▶ $\mathbf{H} = \text{diag}(h(\lambda))$, the **filtered signal** $\mathbf{f}^{\text{filter}}$ is defined as

$$\mathbf{f}^{\text{filter}} = \mathbf{U}\mathbf{H}\mathbf{U}^\top \mathbf{f}.$$

where $\mathbf{U}$ are the **eigenvectors** of $\mathbf{L}$.

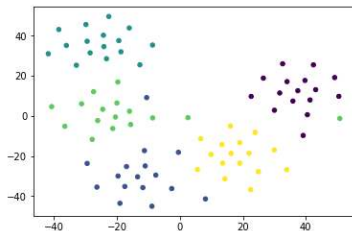
Observations

- To remove **high frequencies** in $\mathbf{f}$, a low-pass filter h **nullifies large values of λ**
- We **substitute $\mathbf{f}$ with $\mathbf{f}^{\text{filter}}$** as an input for the classifier

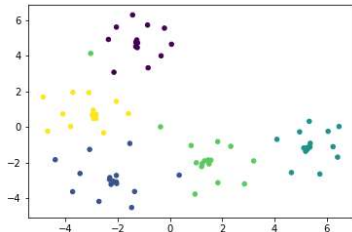
Effect of Low-pass Graph Filters on Centroids (1)

Low-pass filter: $h(\lambda) = 1$ if $\lambda \leq \lambda_m$ and $h(\lambda) = 0$ if not.

- Mini-imageNet data representation before applying any filter



- Mini-imageNet data representation after applying a graph filter



Effect of Low-pass Graph Filters on Centroids (2)

- ▶ $\mathbf{f}_i \sim \mathcal{N}(\boldsymbol{\mu}, \sigma^2 \mathbf{I})$
- ▶ Centroids are computed as

$$\boldsymbol{\gamma} = \frac{1}{K} \sum_{i=1}^K \mathbf{f}_i .$$

- ▶ Filter $\mathbf{H}$ such that $h(\lambda) = 1$ if $\lambda \leq \lambda_m$ and $h(\lambda) = 0$ otherwise ($\lambda \in \boldsymbol{\lambda}$)

Result

- ▶ We build a **complete graph** for each class, and let the number of labeled samples of that class $K \rightarrow \infty$, then

$$\mathbb{E}(\boldsymbol{\gamma}^{\text{filter}}) = \mathbb{E}(\boldsymbol{\gamma}) ,$$

$$\text{Cov}(\boldsymbol{\gamma}^{\text{filter}}) = o(\text{Cov}(\boldsymbol{\gamma})) .$$

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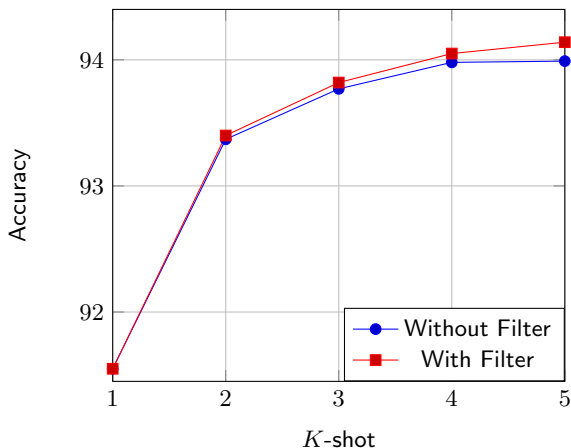
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5-shot Accuracy of the Methodology Compared with State-of-the-art Method, for Various Extractors

- Runs = 100.000
- Number of classes $N = 5$
- Shots $K = 5$
- Queries $q = 15$

Classifier			
Dataset	Extractors	No Filter %	With Filter %
MINet	WRN	88.82 ± 0.013	88.90 ± 0.011
	DNet121	86.82 ± 0.014	86.87 ± 0.014
CUB	WRN	93.99 ± 0.011	94.14 ± 0.009
CIFAR	WRN	90.68 ± 0.015	90.75 ± 0.015
TINet	DNet121	90.44 ± 0.014	90.62 ± 0.013

Evolution of the Accuracy on CUB (extractor: WRN) as a Function of Shots K



Accuracy on the CIFAR-10 Dataset

	CIFAR-10		
Method	WRN	ShakeNet	PyramidNet
original paper	95.82 %	97.96 %	98.56 %
NCM	85.81 %	97.97 %	98.52 %
1-NN	95.81 %	97.95 %	98.54 %
1-NN+Filter	95.92 %	97.97 %	98.61 %

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Conclusion and Perspectives



- A graph-based method is proposed to improve the accuracy of classification methods
- Use of GSP to reduce the noise in the extracted feature vectors
- The effectiveness of the method was shown theoretically
- Gains are obtained in two different settings: few-shot classification and standard classification
- Explore an automatic way of choosing the best filter parameters

Thanks for Your Attention.

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